American Computer Science League

2021-2022 ● Contest 3: Short Problems ● Elementary Division

1. Boolean Algebra

Of the numbers 2, 5, 6, 10, 13, how many satisfy this Boolean expression?

$$NOT\ (x = 10)\ AND\ (x \geq 6)\ OR\ (x = 5)$$

A. 1
B. 2
C. 3
D. 4
E. 5

2. Boolean Algebra

Determine which ordered pair(s) make the following expression FALSE:

~(A~B + ~A~B)

A. (1,1), (1,0)
B. (1,0), (0,1)
C. (1,1), (0,1)
D. (0,0), (0,1)
E. (1,0), (0,0)

3. Boolean Algebra

Determine which two of the following are equivalent Boolean expressions:

a. B(A~B + A) b. B c. A~B + A d. AB

A. a, d
B. a, b
C. a, c
D. c, d
E. b, d

4. Boolean Algebra

Simplify the following Boolean expression:

A~B(A + ~B) + ~AB(~A + B)

A. ~A~B + ~AB
B. A~B + ~AB
C. AB + A~B
D. ~A~B
E. 0

5. Boolean Algebra

Determine which of the following is a tautology:

a. A~B + ~B + A b. ~(A + B) + (A + B)
c. ~A(A~B) + ~B(~AB) d. AB + ~AB + A

A. A~B + ~B + A
B. ~(A + B) + (A + B)
C. ~A(A~B) + ~B(~AB)
D. AB + ~AB + A
E. None of them

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1. 布尔代数

在 2, 5, 6, 10, 13 这些数字中，请问有多少个满足以下布尔表达式？

$$NOT\ (x = 10)\ AND\ (x \geq 6)\ OR\ (x = 5)$$

A. 1
B. 2
C. 3
D. 4
E. 5

2. 布尔代数

判断哪个有序对使下述表达式为假。

~(A~B + ~A~B)

A. (1,1), (1,0)
B. (1,0), (0,1)
C. (1,1), (0,1)
D. (0,0), (0,1)
E. (1,0), (0,0)

3. 布尔代数

判断以下哪两个布尔表达式是等价的。

a. B(A~B + A)　　b. B　　c. A~B + A　　d. AB

A. a, d
B. a, b
C. a, c
D. c, d
E. b, d

4. 布尔代数

化简下方的布尔表达式：

A~B(A + ~B) + ~AB(~A + B)

A. ~A~B + ~AB
B. A~B + ~AB
C. AB + A~B
D. ~A~B
E. 0

5. 布尔代数

以下哪项是永真式（重言式）？

a. A~B + ~B + A　　b. ~(A + B) + (A + B)
c. ~A(A~B) + ~B(~AB)　　d. AB + ~AB + A

A. A~B + ~B + A
B. ~(A + B) + (A + B)
C. ~A(A~B) + ~B(~AB)
D. AB + ~AB + A
E. 均不符合